

Problem 2.18

Define

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Show that

$$a(t) = e^{\int_0^t \delta_r \, dr}.$$

Hint: Notice that

$$\frac{d}{dr}(\ln a(r)) = \delta_r.$$

Solution

Recall that if

$$h(x) = \ln(f(x)),$$

then by the chain rule,

$$h'(x) = \frac{1}{f(x)} f'(x).$$

Therefore, taking $f(r) = a(r)$, we get

$$\frac{d}{dr} \ln(a(r)) = \frac{a'(r)}{a(r)} = \delta_r.$$

Thus,

$$(\ln(a(r)))' = \delta_r.$$

Integrating both sides from 0 to t ,

$$\int_0^t (\ln(a(r)))' \, dr = \int_0^t \delta_r \, dr.$$

Hence,

$$\ln(a(t)) = \int_0^t \delta_r \, dr.$$

Exponentiating both sides, we obtain

$$a(t) = e^{\int_0^t \delta_r \, dr}.$$

Therefore,

$$\boxed{a(t) = e^{\int_0^t \delta_r \, dr}}.$$